HealingMonk

AI Posture & Movement Assessment Engine

AI Technical Reference

How the AI works · body-point detection · measurement maths · accuracy & the on-screen visual layers

This document explains the computer-vision pipeline behind the HealingMonk assessment: the neural pose model used, how the 33 body landmarks are located and marked on the patient, how joint angles and the clinical plumb line are computed, what the accuracy limits are, and what every line and colour on the live overlay means. It is intended for clinicians, reviewers and engineers.

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1 · The AI at a glance

HealingMonk turns an ordinary camera into a clinical-style movement lab. Every measurement runs entirely on the patient's device (in the browser) — no video is uploaded. The engine detects the body, tracks 33 points in 3D in real time, and converts their geometry into clinical measurements (angles, tilts, range-of-motion and a plumb-line alignment score).

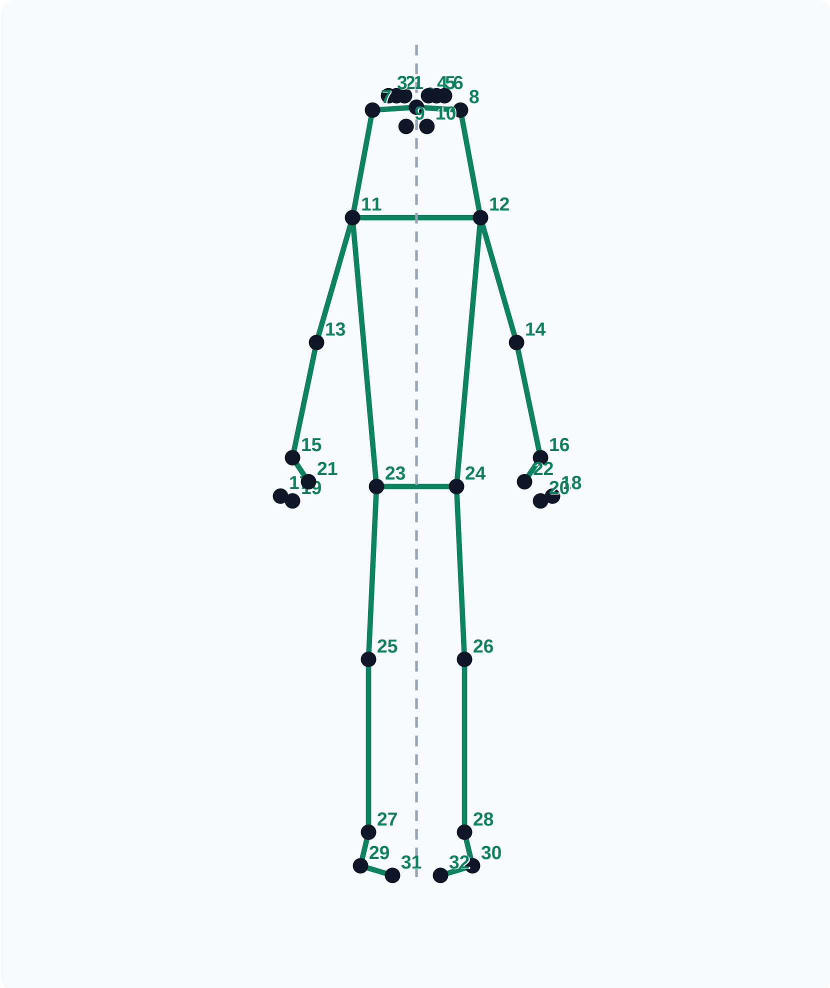
Model	Google MediaPipe Pose Landmarker — "full" variant (float16)
Runtime	@mediapipe/tasks-vision 0.10.35 (WebAssembly + WebGL)
Landmarks	33 full-body points, each with x, y, z (depth) and visibility
Where it runs	Client-side in the browser — camera frames never leave the device
Acceleration	GPU delegate when available, automatic CPU fallback
Mode	VIDEO / real-time, single person (numPoses = 1)
Detection thresholds	pose-detection, presence & tracking confidence all "e 0.60
Smoothing	One-Euro adaptive filter on every point (x, y, z)
Clinical library	34 assessments across posture, upper limb, trunk, lower limb, seated & lying

Important: the automatic numbers are camera-based estimates, not a medical diagnosis. The treating doctor reviews every finding and can override the score — that clinical judgement is the record.

2 · How the body points are detected & marked

The pose model is a two-stage convolutional neural network. A detector first finds the person in the frame; a landmark network then regresses the pixel location of 33 predefined anatomical points, plus a depth value (z) and a visibility confidence (0-1) for each. HealingMonk uses the "full" model, which tracks limbs and depth markedly better than the "lite" model while still running in real time.

The 33 landmarks



0-10	Face / head
11-22	Shoulders, arms, hands
23-28	Hips, knees, ankles
29-32	Heels & toes
- - -	Plumb reference line

Each of the 33 points carries x, y, z (depth) and a 0-1 visibility score. Points below 0.5 visibility are ignored so a hidden limb never corrupts a measurement.

Every point is returned in normalized coordinates (0-1 across the frame), so the maths is resolution-independent. On the live view each point is drawn as a green dot and the joints are joined into a skeleton; the same overlay is "baked into" the captured photo so the doctor sees exactly what the model measured.

All 33 points — where each is marked & what it measures

The complete list below maps every landmark index to its name, the body part it is marked on, and how HealingMonk uses it. Use this as the quick reference when explaining the system.

#	Point name	Body part	Hindi (naam)	Marked on / used for
0	**nose**	Head	**Naak**	Head centre — forward-head check & side-view facing direction
1	**left_eye_inner**	Head	**Baayi aankh (andar)**	Eye landmark — helps estimate vertex (top of head)
2	**left_eye**	Head	**Baayi aankh**	Left eye — head tilt & vertex estimate
3	**left_eye_outer**	Head	**Baayi aankh (bahar)**	Outer eye corner
4	**right_eye_inner**	Head	**Daayi aankh (andar)**	Eye landmark
5	**right_eye**	Head	**Daayi aankh**	Right eye — head tilt & vertex estimate
6	**right_eye_outer**	Head	**Daayi aankh (bahar)**	Outer eye corner
7	**left_ear**	Head	**Bayan kaan**	Left ear (EAM) — craniovertebral angle & side plumb ear point
8	**right_ear**	Head	**Dayan kaan**	Right ear (EAM) — craniovertebral angle & side plumb
9	**mouth_left**	Head	**Munh (bayan)**	Mouth corner — used to estimate the chin point
10	**mouth_right**	Head	**Munh (dayan)**	Mouth corner — used to estimate the chin point
11	**left_shoulder**	Shoulder / torso	**Bayan kandha**	Acromion — shoulder level, plumb line, arm-ROM origin
12	**right_shoulder**	Shoulder / torso	**Dayan kandha**	Acromion — shoulder level, plumb line, arm-ROM origin
13	**left_elbow**	Arm	**Baayi kohni**	Elbow — elbow flexion & shoulder ROM vertex

#	Point name	Body part	Hindi (naam)	Marked on / used for
14	**right_elbow**	Arm	**Daayi kohni**	Elbow — elbow flexion & shoulder ROM vertex
15	**left_wrist**	Arm	**Baayi kalaai**	Wrist — upper-limb ROM endpoint
16	**right_wrist**	Arm	**Daayi kalaai**	Wrist — upper-limb ROM endpoint
17	**left_pinky**	Hand	**Baayi choti ungli**	Little-finger knuckle
18	**right_pinky**	Hand	**Daayi choti ungli**	Little-finger knuckle
19	**left_index**	Hand	**Baayi tarjani**	Index finger — reach / hand endpoint
20	**right_index**	Hand	**Daayi tarjani**	Index finger — reach / hand endpoint
21	**left_thumb**	Hand	**Bayan angutha**	Thumb
22	**right_thumb**	Hand	**Dayan angutha**	Thumb
23	**left_hip**	Pelvis	**Bayan koolha**	Hip / ASIS — pelvic tilt, trunk & hip ROM, plumb line
24	**right_hip**	Pelvis	**Dayan koolha**	Hip / ASIS — pelvic tilt, trunk & hip ROM, plumb line
25	**left_knee**	Leg	**Bayan ghutna**	Knee — knee alignment/flexion, squat depth, plumb line
26	**right_knee**	Leg	**Dayan ghutna**	Knee — knee alignment/flexion, squat depth, plumb line
27	**left_ankle**	Leg	**Bayan takhna**	Ankle (lateral malleolus) — plumb base & dorsiflexion
28	**right_ankle**	Leg	**Dayan takhna**	Ankle (lateral malleolus) — plumb base & dorsiflexion
29	**left_heel**	Foot	**Baayi edi**	Heel — foot alignment & ankle ROM
30	**right_heel**	Foot	**Daayi edi**	Heel — foot alignment & ankle ROM
31	**left_foot_index**	Foot	**Bayan pair ka angutha**	Toe — foot direction & dorsiflexion
32	**right_foot_index**	Foot	**Dayan pair ka angutha**	Toe — foot direction & dorsiflexion

Visibility gate: any point whose visibility drops below 0.5 is treated as "not seen" and excluded, so a briefly hidden hand or foot can never pull a measurement to a wrong value. A live detection-quality indicator (visible points / 33) tells the patient when to reposition.

3 · Steadying the signal (why the points don't shimmer)

Raw neural landmarks jitter by a pixel or two every frame even when the subject is perfectly still, which would make the skeleton shimmer and the angles flicker by ±1-2°. HealingMonk filters every landmark axis with a One-Euro adaptive filter (Casiez et al., CHI 2012):

- When the point is slow / still, it smooths hard — removing all jitter so the deviation line sits rock-steady.
- When the point moves fast, it barely smooths — so there is no visible lag when the patient actually moves.

Tuned constants (per source): baseline cutoff 0.9 Hz, speed coefficient ^{2} = of x, y and z is filtered independently; the filter resets when the body leaves the frame so a re-acquired pose never snaps.

4 · How the measurements are computed

Joint angles

Any joint angle is the angle at point B in the triangle A-B-C (e.g. shoulder-elbow-wrist), computed from the 3D vectors:

```
angle = arccos( (BA · BC) / (|BA| × |BC|) ) × 180/À
    BA = A " B          BC = C " B
  · = dot product     |v| = vector length
Result clamped to a valid 0-180° range.
```

A 2D variant (ignoring z) is used where the clinically correct plane is the image plane — e.g. shoulder level or head tilt. Craniovertebral angle, shoulder flexion/abduction, knee flexion, pelvic tilt and trunk ROM are all derived this way and update live as the patient moves.

Clinical plumb line

A plumb line is the vertical reference a physiotherapist drops through the body to screen standing posture. HealingMonk reconstructs it per view (front / side / back) and reports how far each anatomical checkpoint sits off that line, as a percentage of body height — so the score is scale-free.

- Front: central midline through vertex !' nose !' chin !' sternum !' umbilicus !' p
  plus left/right symmetry (shoulders, clavicle, ASIS, knees, feet).
- Side: ear (EAM) !' shoulder (acromion) !' hip (greater trochanter) !' knee !' ar
- Back: occiput !' cervical/thoracolumbar spine !' gluteal cleft !' knee & ankle

MediaPipe only returns 33 surface points, so several true clinical landmarks (vertex, sternum, umbilicus, occiput, greater trochanter) are estimated geometrically from the points that exist. Tolerances: central/sagittal checkpoints within ±4% (front/back) or ±5% (side) of body height; left/right pairs level within 3°.

Severity rating

The plumb score (mean absolute checkpoint offset) is graded into the same four-band scale used across the report:

```
score "d 2%   !'  Normal    (green)
score "d 4%   !'  Mild      (yellow)
score "d 7%   !'  Moderate  (orange)
score  > 7%   !'  Severe    (red)
```

Overall posture health score (report)

The headline 0-100 score on the report is a weighted deviation from ideal across all captured findings:

```
penalty(finding) = 0 normal · 1 mild · 2 moderate · 3 severe
score = 100 "  (£ penalty / (3 × findings)) × 100
clamped to 0-100 and rounded.
```

5 · Accuracy, feasibility & limitations

Across the 34-assessment library each posture is tagged with an AI-feasibility rating that reflects how reliably a single camera can measure it: most posture, upper-limb and trunk screens are rated High; a few depth- or rotation-sensitive measures (e.g. anterior pelvic tilt, spinal rotation) are rated Partial or Medium and are flagged for extra doctor scrutiny.

High feasibility	Movement is clearly visible in the camera plane — reliable estimate (majority of the library).
Medium	Usable, but sensitive to camera angle or subtle motion — verify against exam.
Partial	Depth / rotation limited by a single 2D camera — treat as a screening indicator only.

What affects accuracy in practice

- Lighting — natural or bright even light is best; low light or backlighting reduces landmark confidence.
- Framing & distance — the full body (or the relevant segment) must be in frame; a live quality bar guides repositioning.
- Clothing — contrasting, non-baggy clothing gives the cleanest landmarks.
- Depth ambiguity — a single camera estimates z (depth), so purely front-back movements are the hardest and are rated Partial.

Performance envelope: roughly 50-90 ms per frame end-to-end (detection + angles + rendering), with GPU acceleration when the device allows. Because everything is an estimate, the report carries an explicit AI disclaimer and the doctor's clinical score overrides the automatic value on every finding.

6 · The on-screen visuals (3D points, skeleton & animation)

Several layers are drawn live over the camera feed and preserved in the report so the assessment is fully transparent:

- 3D landmark dots — all visible points from the 33-point model, positioned using the model's x, y and z (depth) output.
- Skeleton overlay — green lines joining the points (shoulders !' elbows !' wri
  the tracked body is obvious at a glance.
- Moving body axis — the patient's actual head!'foot centre line, drawn greer
  where it drifts off, giving a live "correct / off" read.
- Fixed plumb reference — a dashed vertical in the severity colour, with each anatomical checkpoint labelled by its deviation (e.g. "Ear 6% ant").
- Symmetry bars (front view) — dashed left!"right connectors showing should
  high.
- Muscle-map illustration — for each posture the report shows an anatomical figure (anterior or posterior) with the targeted muscle groups highlighted, so the patient sees which area the assessment relates to.

Together these turn a raw camera frame into a readable clinical picture: the patient photo with the AI points baked in, next to the ideal reference — exactly what the doctor scores against.

Appendix · The full assessment library

Every posture, range-of-motion and movement screen the engine can run, with the single-camera measurement it produces and its AI-feasibility rating. Feasibility colour: green = High, purple = Medium, amber = Partial. (34 assessments.)

#	Assessment	Region	Measurement	Feasibility
1	**Full Body — Center (Front)**	Full Body	Overall Alignment (°)	High
2	**Full Body — Back**	Full Body	Overall Alignment (°)	High
3	**Full Body — Left Side**	Full Body	Trunk Vertical Lean (°)	High
4	**Full Body — Right Side**	Full Body	Trunk Vertical Lean (°)	High
5	**Forward Head Posture**	Cervical	Craniovertebral Angle (°)	High
6	**Shoulder Level Symmetry**	Shoulder	Shoulder Tilt Angle (°)	High
7	**Pelvic Level Symmetry**	Hip	Pelvic Tilt Angle (°)	High
8	**Lateral Spinal Alignment**	Thoracic	Trunk Lateral Deviation (°)	Partial
9	**Lateral Head Tilt**	Cervical	Head Tilt Angle (°)	High
10	**Knee Alignment (Valgus/Varus)**	Knee	Knee Deviation Angle (°)	High
11	**Shoulder Flexion ROM (Overhead Reach)**	Shoulder	Shoulder Flexion Angle (°)	High
12	**Thoracic Kyphosis (Upper-Back Rounding)**	Thoracic	Upper-Trunk Rounding (°)	Partial
13	**Anterior Pelvic Tilt**	Pelvis	Trunk–Thigh Tilt (°)	Partial
14	**Side Plank (concave side)**	Spine	Trunk Asymmetry (°)	Partial
15	**Squat Depth (Knee Flexion)**	Knee	Knee Flexion (°)	High
16	**Hip Abduction ROM (Side Leg Raise)**	Hip	Hip Abduction Angle (°)	High
17	**Shoulder Abduction ROM (Side Arm Raise)**	Shoulder	Shoulder Abduction Angle (°)	High
18	**Elbow Flexion ROM**	Elbow	Elbow Flexion (°)	High
19	**Trunk Forward Flexion (Toe Touch)**	Spine	Forward Bend Angle (°)	High
20	**Trunk Lateral Flexion (Side Bend)**	Spine	Side Bend Angle (°)	High
21	**Cervical Flexion (Chin to Chest)**	Cervical	Neck Flexion Angle (°)	Medium
22	**Cervical Extension (Look Up)**	Cervical	Neck Extension Angle (°)	Medium
23	**Shoulder Extension ROM (Arm Back)**	Shoulder	Shoulder Extension Angle (°)	High
24	**Hip Flexion ROM (Knee Raise)**	Hip	Hip Flexion Angle (°)	High
25	**Active Knee Flexion (Heel to Buttock)**	Knee	Knee Flexion (°)	High
26	**Forward Lunge Depth**	Knee	Front Knee Flexion (°)	High
27	**Ankle Dorsiflexion (Knee to Wall)**	Ankle	Ankle Angle (°)	Medium
28	**Single-Leg Stance (Pelvic Drop)**	Hip	Pelvic Drop (°)	High
29	**Overhead Squat (Knee Control)**	Knee	Knee Valgus Deviation (°)	High
30	**Standing Back Extension**	Spine	Back Extension Angle (°)	Medium

#	Assessment	Region	Measurement	Feasibility
31	**Seated Forward Head (Sitting)**	Cervical	Seated Craniovertebral Angle (°)	High
32	**Seated Slump / Upper-Back Rounding (Sitting)**	Thoracic	Seated Trunk Rounding (°)	Partial
33	**Straight Leg Raise — SLR (Lying)**	Hip	SLR Hip Flexion (°)	High
34	**Knee-to-Chest Hip Flexion (Lying)**	Hip	Hip Flexion (knee bent) (°)	High